

MURALIDHAR REDDY ANUMANDLA

Data Engineer | SQL • Python • ETL/ELT • Data Warehousing • Git

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PROFESSIONAL SUMMARY

Re-engineered a 140M-row Spark job to run 5-10x faster and cut reporting query time ~30% — Data Engineer with over 1 year of production experience across the pipeline lifecycle in Azure, Databricks, PySpark/Python, and advanced SQL (joins, window functions, optimization, indexing). Owns 2 end-to-end ETL/ELT pipelines and monitors 25+ more for SLA adherence on regulated healthcare data, building governed dimensional data warehouse models. M.S. Computer Science, VCU (GPA 3.90/4.0, Dec 2025).

TECHNICAL SKILLS

Core: Advanced SQL (joins, window functions, query optimization, indexing), Python, Pandas, PySpark, Spark SQL, Apache Spark, Databricks, Delta Lake, Unity Catalog, Azure Data Factory, dbt, Apache Airflow

Cloud & Platforms: Microsoft Azure, Azure Blob Storage, Azure SQL Database, Azure Synapse, Azure Key Vault

Data Engineering: ETL/ELT pipeline design, distributed data processing (Spark), incremental loads, event-driven batch ingestion (Kafka), upsert/merge logic, data warehouse modeling (star schema), partitioning, pipeline monitoring & alerting

Tools & Practices: Git (PR-based code review), CI/CD (GitHub Actions), Linux, runbooks & documentation, root-cause debugging, Agile/Scrum, Power BI

PROFESSIONAL EXPERIENCE

Appex Systems (Client: UnitedHealth Group) — Data Engineer

Full-time Data Engineer | Feb 2026 – Present

Data Engineer Intern | Jun 2025 – Dec 2025

- Own 2 end-to-end ETL/ELT pipelines (Azure Data Factory + Databricks) spanning ingestion, transformation, and load, while monitoring 25+ additional production pipelines for health, failures, and SLA adherence across daily, hourly, and monthly schedules.
- Re-engineered a Spark job failing after 20 hours on ~**140M rows** with an OOM error via window-function dedup, early filtering, Parquet staging, and removing a redundant write — cutting runtime to 2–4 hours (**~5–10x faster**).
- Built a Kafka-fed subscriber-preference pipeline into silver Delta tables, decrypting PII via Azure Key Vault and applying business rules to publish consolidated unsub flags per line of business for downstream email-suppression compliance.
- Led the one-time migration of ~**3.6M legacy all-or-none unsubscribe records** into the category-level model to retire the legacy tables.
- Root-caused a duplicate-record defect in the Coverage table of a Kafka-fed membership pipeline (**8 tables**, continuous production) with the BI/analytics team and a solutions architect — traced to a wrong merge key and a faulty derived column, resolving both and restoring accurate member-level reporting for stakeholders.
- Enforced onshore/offshore access restrictions on regulated healthcare data by applying data-classification tags to sensitive tables, maintaining governed Delta tables in Unity Catalog daily.

PERSONAL PROJECTS

Multi-Source Batch ETL & Data Warehouse on Databricks 

Azure Databricks — PySpark — Delta Lake — Azure Data Factory — SQL

- Built a production-grade Medallion (Bronze/Silver/Gold) lakehouse in PySpark, unifying 4-6 daily CSV feeds and operational SQL tables into a pipeline processing **120K+ sales records/day** into a governed, Gold-layer star-schema warehouse, with automated data-quality checks catching null, type, and duplicate errors before they reach Gold.
- Replicated real production reliability patterns solo: a Delta MERGE upsert that correctly deduped 6,583 duplicate transaction IDs across a 660K-row, 5-day backfill, plus automated daily orchestration, retry-with-backoff, and failure alerting.
- Benchmarked partitioning and Z-ordering against an unoptimized baseline, cutting query latency **40-60%**, and layered in Delta Lake time-travel for point-in-time rollback and full data versioning.

Orders Lakehouse: ADF + Delta Lake + dbt + Airflow 

Azure Data Factory — Delta Lake — dbt — Airflow

- Engineered an idempotent, failure-safe Azure Data Factory pipeline (Lookup + Until-loop pagination) ingesting a paginated REST API into a Delta Lake bronze layer, advancing the watermark **only on full success** so partial failures never corrupt state — validated by an automated test suite covering interruption/retry scenarios.
- Modeled a governed dbt star-schema mart (fct_orders + 3 dimension tables) on the bronze layer, enforcing data integrity with 19 automated tests (not_null/unique/relationships) and auto-generated lineage documentation.
- Orchestrated the full dbt run via a 4-stage **Airflow DAG** (sensor → ingest → run → test), decoupled from ADF's own ingestion trigger to keep transformation and ingestion independently schedulable.

E-Commerce Sales Analytics — Power BI Dashboard (Star Schema, DAX) 

- Modeled a 5-table Power BI star schema (**932K+ transactions, \$324M revenue**), 6 DAX measures, 4 visuals; benchmarked vs. a flat table in DuckDB.

EDUCATION

M.S. in Computer Science | **Virginia Commonwealth University** | GPA 3.90/4.0 | 2024 – Dec 2025

B.Tech in Computer Science and Business Systems | **GRIET** | GPA 8.31/10 | 2020 – 2024